

20261007 Quiz

for (MATH200) Differential Equations

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Problem 3.5.4. Find the general solution of the differential equation

$$y'' - 2y' - 3y = -6te^{-t}.$$

Solution. The corresponding homogeneous equation has characteristic equation

$$r^2 - 2r - 3 = (r - 3)(r + 1) = 0,$$

so

$$y_h = c_1 e^{3t} + c_2 e^{-t}.$$

The forcing term is a degree-one polynomial times e^{-t} . Since $r = -1$ is a simple characteristic root, the usual trial $(At + B)e^{-t}$ overlaps with the homogeneous solution and must be multiplied by t . Thus set

$$y_p = t(At + B)e^{-t}.$$

Substitution gives

$$y_p'' - 2y_p' - 3y_p = (-8At + 2A - 4B)e^{-t}.$$

Matching coefficients with $-6te^{-t}$ yields

$$-8A = -6, \quad 2A - 4B = 0,$$

so

$$A = \frac{3}{4}, \quad B = \frac{3}{8}.$$

Therefore

$$y = c_1 e^{3t} + c_2 e^{-t} + \frac{3}{8}te^{-t} + \frac{3}{4}t^2e^{-t}.$$

□

Problem 3.5.5. Find the general solution of the differential equation

$$y'' + 2y' = 5 + 4\sin(2t).$$

Solution. The characteristic equation of the corresponding homogeneous equation is

$$r^2 + 2r = r(r + 2) = 0,$$

so

$$y_h = c_1 + c_2 e^{-2t}.$$

Because the forcing term is a sum, we choose a trial particular solution by treating the constant and trigonometric terms together. A constant trial would duplicate the homogeneous solution 1, so the constant forcing requires a factor of t . The trigonometric forcing has no such duplication. Hence let

$$y_p = At + B \cos(2t) + C \sin(2t).$$

Then

$$y_p'' + 2y_p' = 2A + (4C - 4B) \cos(2t) + (-4B - 4C) \sin(2t).$$

Matching coefficients with $5 + 4 \sin(2t)$ gives

$$2A = 5, \quad C - B = 0, \quad -B - C = 1.$$

Thus

$$A = \frac{5}{2}, \quad B = C = -\frac{1}{2}.$$

Therefore

$$y = c_1 + c_2 e^{-2t} + \frac{5}{2}t - \frac{1}{2} \sin(2t) - \frac{1}{2} \cos(2t).$$

□

Problem 3.5.6. Find the general solution of the differential equation

$$y'' + 2y' + y = 4e^{-t}.$$

Solution. The characteristic equation is

$$r^2 + 2r + 1 = (r + 1)^2 = 0,$$

so

$$y_h = (c_1 + c_2 t)e^{-t}.$$

The forcing term is proportional to e^{-t} , and $r = -1$ is a characteristic root of multiplicity two. Therefore the trial must be multiplied by t^2 :

$$y_p = At^2 e^{-t}.$$

A direct substitution gives

$$y_p'' + 2y_p' + y_p = 2Ae^{-t}.$$

Hence $2A = 4$, so $A = 2$. Therefore

$$y = c_1 e^{-t} + c_2 t e^{-t} + 2t^2 e^{-t}.$$

□

Problem 3.5.16. Determine a suitable form for $Y(t)$ if the method of undetermined coefficients is to be used for

$$y'' + 3y' = 2t^4 + t^2 e^{-3t} + 2 \sin(3t).$$

Solution. The characteristic equation of the corresponding homogeneous equation is

$$r^2 + 3r = r(r + 3) = 0,$$

with roots 0 and -3 .

For the polynomial forcing $2t^4$, the usual degree-four polynomial trial overlaps with the homogeneous solution corresponding to $r = 0$, so it must be multiplied by t . For the term t^2e^{-3t} , the exponential factor corresponds to the simple root $r = -3$, so this trial also requires one factor of t . The term $2\sin(3t)$ has no duplication with the homogeneous solution and requires both sine and cosine terms. Thus a suitable form is

$$\begin{aligned} Y(t) = & t(A_0t^4 + A_1t^3 + A_2t^2 + A_3t + A_4) \\ & + t(B_0t^2 + B_1t + B_2)e^{-3t} \\ & + D\sin(3t) + E\cos(3t). \end{aligned}$$

□

Problem 3.5.18. Determine a suitable form for $Y(t)$ if the method of undetermined coefficients is to be used for

$$y'' + 2y' + 2y = 2e^{-t} + 2e^{-t}\cos t + 4e^{-t}t^2\sin t.$$

Solution. The characteristic equation of the corresponding homogeneous equation is

$$r^2 + 2r + 2 = 0,$$

with roots $-1 \pm i$. Thus $e^{-t}\cos t$ and $e^{-t}\sin t$ belong to the homogeneous solution family.

The forcing term $2e^{-t}$ alone has no duplication with the homogeneous solution, so it contributes Ae^{-t} . The remaining two forcing terms belong to the common family $e^{-t}P_2(t)\cos t$ and $e^{-t}P_2(t)\sin t$. Since differentiation mixes the sine and cosine terms, both degree-two polynomial factors must be included. Because $-1 \pm i$ are simple characteristic roots, the entire trigonometric trial must also be multiplied by t . Hence a suitable form is

$$\begin{aligned} Y(t) = & Ae^{-t} \\ & + t(B_0t^2 + B_1t + B_2)e^{-t}\cos t \\ & + t(D_0t^2 + D_1t + D_2)e^{-t}\sin t. \end{aligned}$$

□

Problem 3.5.21. Determine a suitable form for $Y(t)$ if the method of undetermined coefficients is to be used for

$$y'' - 4y' + 4y = 4t^2 + 4te^{2t} + t\sin(2t).$$

Solution. The characteristic equation of the corresponding homogeneous equation is

$$r^2 - 4r + 4 = (r - 2)^2 = 0,$$

so $r = 2$ is a double root.

The polynomial forcing $4t^2$ requires a general quadratic polynomial. For $4te^{2t}$, the usual trial $e^{2t}(B_0t + B_1)$ duplicates the homogeneous solution with multiplicity two, so it must be

multiplied by t^2 . Finally, the forcing $t \sin(2t)$ requires both sine and cosine terms with degree-one polynomial coefficients, and there is no resonance at $\pm 2i$. Therefore a suitable form is

$$\begin{aligned} Y(t) = & A_0 t^2 + A_1 t + A_2 \\ & + t^2(B_0 t + B_1)e^{2t} \\ & + (D_0 t + D_1) \sin(2t) \\ & + (E_0 t + E_1) \cos(2t). \end{aligned}$$

□

Problem 3.6.2. Use the method of variation of parameters to find a particular solution of

$$y'' - y' - 2y = 4e^{-t}.$$

Then check your answer by using the method of undetermined coefficients.

Solution. The corresponding homogeneous equation has characteristic equation

$$r^2 - r - 2 = (r + 1)(r - 2) = 0.$$

Choose the fundamental solutions

$$y_1 = e^{-t}, \quad y_2 = e^{2t}.$$

Their Wronskian is

$$W(y_1, y_2)(t) = \begin{vmatrix} e^{-t} & e^{2t} \\ -e^{-t} & 2e^{2t} \end{vmatrix} = 3e^t.$$

For variation of parameters, write

$$y_p = u_1 y_1 + u_2 y_2.$$

Since $g(t) = 4e^{-t}$,

$$u_1' = -\frac{y_2 g}{W(y_1, y_2)} = -\frac{4}{3}, \quad u_2' = \frac{y_1 g}{W(y_1, y_2)} = \frac{4}{3}e^{-3t}.$$

Taking convenient antiderivatives,

$$u_1 = -\frac{4}{3}t, \quad u_2 = -\frac{4}{9}e^{-3t}.$$

Thus

$$u_1 y_1 + u_2 y_2 = -\frac{4}{3}te^{-t} - \frac{4}{9}e^{-t}.$$

The second term is a homogeneous solution, so it may be absorbed into y_h . Hence we may take

$$y_p = -\frac{4}{3}te^{-t}.$$

To check by undetermined coefficients, note that e^{-t} is already a homogeneous solution and $r = -1$ is a simple characteristic root. Therefore use the resonant trial

$$y_p = Ate^{-t}.$$

Substitution gives

$$y_p'' - y_p' - 2y_p = -3Ae^{-t}.$$

Thus $-3A = 4$, so $A = -4/3$, confirming the result.

□

Problem 3.6.11. Verify that

$$y_1(t) = t, \quad y_2(t) = te^t$$

satisfy the corresponding homogeneous equation; then find a particular solution of

$$t^2 y'' - t(t+2)y' + (t+2)y = 6t^3, \quad t > 0.$$

Solution. For $y_1 = t$, we have $y_1' = 1$ and $y_1'' = 0$, so

$$t^2 y_1'' - t(t+2)y_1' + (t+2)y_1 = -t(t+2) + t(t+2) = 0.$$

For $y_2 = te^t$,

$$y_2' = (t+1)e^t, \quad y_2'' = (t+2)e^t,$$

and hence

$$\begin{aligned} t^2 y_2'' - t(t+2)y_2' + (t+2)y_2 \\ = t^2(t+2)e^t - t(t+2)(t+1)e^t + t(t+2)e^t = 0. \end{aligned}$$

Thus both functions solve the homogeneous equation.

Before applying variation of parameters, divide the differential equation by t^2 ; this is valid because $t > 0$. The standard form is

$$y'' - \frac{t+2}{t}y' + \frac{t+2}{t^2}y = 6t,$$

so the forcing term in the variation-of-parameters formula is $g(t) = 6t$. The Wronskian is

$$\begin{aligned} W(y_1, y_2)(t) &= \begin{vmatrix} t & te^t \\ 1 & (t+1)e^t \end{vmatrix} \\ &= t^2 e^t, \end{aligned}$$

which is nonzero for $t > 0$. Therefore y_1 and y_2 form a fundamental set on this interval.

With $y_p = u_1 y_1 + u_2 y_2$,

$$u_1' = -\frac{y_2 g}{W(y_1, y_2)} = -6, \quad u_2' = \frac{y_1 g}{W(y_1, y_2)} = 6e^{-t}.$$

Taking antiderivatives,

$$u_1 = -6t, \quad u_2 = -6e^{-t}.$$

This gives

$$y_p = -6t^2 - 6t.$$

Particular solutions are not unique: the term $-6t$ is a homogeneous solution because $y_1 = t$. Absorbing it into the homogeneous part gives the simpler particular solution

$$\boxed{y_p = -6t^2}.$$

□

Problem 7.1.3. Transform the differential equation

$$u^{(4)} - 3u = 0$$

into a system of first-order equations.

Solution. Introduce one state variable for each derivative up to order three:

$$x_1 = u, \quad x_2 = u', \quad x_3 = u'', \quad x_4 = u''.$$

This choice makes each derivative of a state variable equal to the next state variable, except for the last equation, which is supplied by the original differential equation. Thus

$$\begin{cases} x_1' = x_2, \\ x_2' = x_3, \\ x_3' = x_4, \\ x_4' = 3x_1. \end{cases}$$

Equivalently, with

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix},$$

the system can be written in state-vector form as

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 3 & 0 & 0 & 0 \end{pmatrix} \mathbf{x}.$$

□

Problem 7.1.4. Transform the initial value problem

$$u'' + 2u' + 4u = 2 \cos(3t), \quad u(0) = 1, \quad u'(0) = -2$$

into an initial value problem for two first-order equations.

Solution. Let

$$x_1 = u, \quad x_2 = u'.$$

Then $x_1' = x_2$, while the original differential equation gives

$$x_2' = u'' = -4u - 2u' + 2 \cos(3t) = -4x_1 - 2x_2 + 2 \cos(3t).$$

The initial conditions become $x_1(0) = 1$ and $x_2(0) = -2$. Hence the required first-order initial value problem is

$$\begin{cases} x_1' = x_2, \\ x_2' = -4x_1 - 2x_2 + 2 \cos(3t), \\ x_1(0) = 1, \quad x_2(0) = -2. \end{cases}$$

In state-vector form,

$$\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -4 & -2 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 0 \\ 2 \cos(3t) \end{pmatrix}, \quad \mathbf{x}(0) = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

□

Problem 7.4.2. Consider the homogeneous system

$$\mathbf{x}' = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} \mathbf{x}$$

and the vector-valued functions

$$\mathbf{x}^{(1)}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t, \quad \mathbf{x}^{(2)}(t) = \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-t}.$$

- (a) Show that the given functions are solutions of the given system of differential equations.
- (b) Show that $\mathbf{x} = c_1 \mathbf{x}^{(1)} + c_2 \mathbf{x}^{(2)}$ is also a solution of the given system for any values of c_1 and c_2 .
- (c) Show that the given functions form a fundamental set of solutions of the given system.
- (d) Find the solution of the given system that satisfies the initial condition

$$\mathbf{x}(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

- (e) Find $W(\mathbf{x}^{(1)}, \mathbf{x}^{(2)})(t)$.
- (f) Show that the Wronskian $W = W(\mathbf{x}^{(1)}, \mathbf{x}^{(2)})$ found in part (e) is a solution of Abel's equation

$$W' = (p_{11}(t) + p_{22}(t))W.$$

Solution. Let

$$\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}.$$

- (a) For the first vector,

$$\mathbf{A} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

so

$$\mathbf{A}\mathbf{x}^{(1)}(t) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t = (\mathbf{x}^{(1)})'(t).$$

For the second vector,

$$\mathbf{A} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = - \begin{pmatrix} 1 \\ 3 \end{pmatrix},$$

and therefore

$$\mathbf{A}\mathbf{x}^{(2)}(t) = - \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-t} = (\mathbf{x}^{(2)})'(t).$$

Thus both functions are solutions of $\mathbf{x}' = \mathbf{A}\mathbf{x}$.

- (b) By linearity,

$$\begin{aligned} \mathbf{x}' &= c_1 (\mathbf{x}^{(1)})' + c_2 (\mathbf{x}^{(2)})' \\ &= c_1 \mathbf{A}\mathbf{x}^{(1)} + c_2 \mathbf{A}\mathbf{x}^{(2)} \\ &= \mathbf{A}(c_1 \mathbf{x}^{(1)} + c_2 \mathbf{x}^{(2)}) \\ &= \mathbf{A}\mathbf{x}. \end{aligned}$$

Hence every linear combination $c_1 \mathbf{x}^{(1)} + c_2 \mathbf{x}^{(2)}$ is also a solution.

(c) Their Wronskian is

$$\begin{aligned} W(\mathbf{x}^{(1)}, \mathbf{x}^{(2)})(t) &= \det \begin{pmatrix} e^t & e^{-t} \\ e^t & 3e^{-t} \end{pmatrix} \\ &= 3 - 1 = 2. \end{aligned}$$

Since the Wronskian is nonzero, the two solutions are linearly independent and therefore form a fundamental set of solutions.

(d) Since the two functions form a fundamental set, write

$$\mathbf{x}(t) = c_1 \mathbf{x}^{(1)}(t) + c_2 \mathbf{x}^{(2)}(t).$$

At $t = 0$, the initial condition gives

$$c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

so

$$c_1 + c_2 = 1, \quad c_1 + 3c_2 = 2.$$

Thus

$$c_1 = c_2 = \frac{1}{2},$$

and

$$\boxed{\mathbf{x}(t) = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + \frac{1}{2} \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-t}}.$$

(e) From part (c),

$$\boxed{W(\mathbf{x}^{(1)}, \mathbf{x}^{(2)})(t) = 2}.$$

(f) Here

$$p_{11}(t) + p_{22}(t) = 2 + (-2) = 0 = \text{tr}(\mathbf{A}).$$

Since $W(t) = 2$ is constant,

$$W'(t) = 0 = (p_{11}(t) + p_{22}(t))W(t).$$

Thus the Wronskian satisfies Abel's equation.

□